Ratio & Proportion

Basics of Ratio

Q1. Simplify 24 : 36
A) 2:3 B) 3:4 C) 4:5 D) 5:6

Q2. Ratio of 0.5 : 2
A) 1:2 B) 1:3 C) 1:4 D) 2:3

Q3. 18 : 24 = ?
A) 2:3 B) 3:4 C) 4:5 D) 5:6

Q4. Ratio of 45 min to 1 hour
A) 3:4 B) 4:5 C) 2:3 D) 5:6

Ratio Expressions

Q5. If a:b = 3:5 → (2a+b):(a+2b)?
A) 11:13 B) 13:11 C) 7:8 D) 8:7

Q6. If x:y = 4:5 → (x+y):(x−y)?
A) 9:1 B) 1:9 C) 5:4 D) 4:1

Q7. If a:b = 3:4 → (2a+3b):(3a+2b)?
A) 17:18 B) 18:17 C) 15:14 D) 14:15

Equivalent Ratios

Q8. Equivalent to 4 : 7
A) 12:21 B) 8:10 C) 16:20 D) 10:14

Q9. If a:b = 2:3 → (3a):(3b)?
A) 2:3 B) 3:2 C) 6:9 D) Both A & C

4. Proportion

Q10. If 4:5 = x:20, find x
A) 12 B) 15 C) 16 D) 18

Q11. If x/3 = 12/16, find x
A) 2.25 B) 3 C) 4 D) 5

Q12. If a:b = c:d, then
A) ad = bc B) ab = cd C) a+b=c+d D) none

5. Direct & Inverse Proportion

Q13. 5 men → 10 days, 10 men → ?
A) 2 B) 5 C) 10 D) 20

Q14. 8 men → 12 days, men for 6 days?
A) 12 B) 14 C) 16 D) 18

Q15. 6 workers earn ₹1200 → 9 workers?
A) 1500 B) 1800 C) 2000 D) 2200

6. Compound & Duplicate Ratio

Q16. Compound ratio of (2:3) & (4:5)
A) 6:15 B) 8:15 C) 2:5 D) 4:7

Q17. Duplicate ratio of 3:5
A) 6:10 B) 9:25 C) 3:25 D) 9:10

Q18. Triplicate ratio of 2:3
A) 6:9 B) 8:27 C) 4:9 D) 12:27

7. Partnership

Q19. A ₹5000, B ₹8000 → profit ratio?
A) 5:8 B) 8:5 C) 13:5 D) 5:13

Q20. Equal money, A 6 months, B 12 months
A) 1:2 B) 2:1 C) 1:1 D) 3:2

Q21. A ₹6000 (8m), B ₹8000 (6m)
A) 4:3 B) 3:2 C) 2:1 D) 1:1

8. Mixtures

Q22. Milk:Water = 5:3, total 16L → milk?
A) 8 B) 10 C) 12 D) 14

Q23. 20L mixture, 25% water → water?
A) 5 B) 10 C) 15 D) 8

Q24. 40% of 50L → alcohol?
A) 10 B) 20 C) 25 D) 30

9. Ages

Q25. A:B = 3:4 → after 5 yrs = 4:5 → A?
A) 15 B) 20 C) 25 D) 30

Q26. A:B = 5:7 → after 10 yrs = 7:9 → A?
A) 25 B) 30 C) 35 D) 40

10. Multi Ratio

Q27. a:b = 2:3, b:c = 4:5 → a:b:c?
A) 8:12:15 B) 2:3:5 C) 4:6:5 D) 6:9:10

Q28. a:b = 2:3, b:c = 5:7 → a:c?
A) 10:21 B) 2:7 C) 6:7 D) 4:7

11. Ratio Division

Q29. Divide ₹840 in 2:3:7
A) 120,180,420
B) 140,210,490
C) 160,240,440
D) 100,200,540

Q30. Divide ₹900 in 1:2:3
A) 100,200,600
B) 150,300,450
C) 200,300,400
D) 250,300,350

Advanced

Q31. Ratio 7:9, sum 160 → smaller?
A) 60 B) 70 C) 80 D) 90

Q32. If 3x = 5y → x:y
A) 3:5 B) 5:3 C) 1:1 D) 2:3

Q33. If x:y = 3:4 → x^2:y^2
A) 9:16 B) 3:4 C) 6:8 D) 12:16

Q34. 20% as ratio
A) 1:4 B) 2:5 C) 1:5 D) 4:5

Mixed / Trick Questions

Q35. Which is greatest?
A) 5:6 B) 7:8 C) 9:10 D) Equal

Q36. a:b:c = 3:4:5, sum = 120 → c?
A) 30 B) 40 C) 50 D) 60

Q37. Income ratio 4:5, expenditure 3:4 → savings?
A) 2:1 B) 3:2 C) 1:2 D) Cannot determine

14. Extra

Q38. If 2x:3x = 8:12
A) 1 B) 2 C) 3 D) Cannot determine

Q39. Salary increases: A +20%, B +10% → new ratio (2:3)?
A) 4:5 B) 8:9 C) 6:7 D) 5:6

Q40. 30L mixture 2:1 → make 1:1 → water added?
A) 5 B) 10 C) 15 D) 20

Answer Key

1-A
2-C
3-B
4-A
5-A
6-A

7-A
8-A
9-D
10-C
11-A
12-A
13-B
14-C
15-B
16-B
17-B
18-B
19-A
20-A
21-A
22-B
23-A
24-B
25-A
26-B
27-A
28-A
29-B
30-B
31-B
32-B
33-A
34-A
35-D
36-C
37-D
38-D
39-B
40-C